been unable for some time to address the question of what properties any one of the string theories would have if its coupling constant should be larger than 1—the so-called strongly coupled behavior. The claim of Witten and others, as we now discuss, is that this crucial question can now be answered. Their results convincingly suggest that, together with a sixth theory we have yet to describe, the strong coupling behavior of any of these theories has a dual description in terms of the weak coupling behavior of another, and vice versa.

To gain a more tangible sense of what this means, you might want to keep the following analogy in mind. Imagine two rather sheltered individuals. One loves ice but, strangely enough, has never seen water (in its liquid form). The other loves water but, equally strangely, has never seen ice. Through a chance meeting, they decide to team up for a camping trip in the desert. When they set out to leave, each is fascinated by the other's gear. The ice-lover is captivated by the water-lover's silky smooth transparent liquid, and the water-lover is strangely drawn to the remarkable solid crystal cubes brought by the ice-lover. Neither has any inkling that there is actually a deep relationship between water and ice; to them, they are two completely different substances. But as they head out into the scorching heat of the desert, they are shocked to find that the ice slowly begins to turn into water. And, in the frigid cold of the desert night, they are equally shocked to find that the liquid water slowly begins to turn into solid ice. They realize that these two substances—which they initially thought to be completely unrelated—are intimately connected.

The duality among the five string theories is somewhat similar: Roughly speaking, the string coupling constants play a role analogous to temperature in our desert analogy. Like ice and water, any two of the five string theories, at first sight, appear to be completely distinct. But as we vary their respective coupling constants, the theories transmute among themselves. Just as ice transmutes into water as we raise its temperature, one string theory can transmute into another as we increase the value of its coupling constant. This takes us a long way toward showing that all of the string theories are dual descriptions of one single underlying structure—the analog of H2O for water and ice.

The reasoning underlying these results relies almost entirely on the use of arguments rooted in principles of symmetry Let's discuss this.